

Areas Related to Circles

Chapter 11 - Hinglish Notes, Area, Sector, Segment + Solutions

Ye Chapter 11 (Areas Related to Circles) ka complete guide hai. Pehle red-page ke core basics (circle parts, pi, area vs perimeter, sector, angle, fraction of circle), fir chapter ka main content - circumference, area, sector aur segment ke formulas. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Circle - Radius, Diameter

EXAM * Exam me CONSTANTLY aata hai

- Radius (r) = centre se circle tak ki doori.
- Diameter = $2r$ (circle ke aar-paar, centre se hokar).

2. pi (PI) Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

- π = circumference / diameter = 3.14 (ya $22/7$).
- Radius 7 ka multiple ho to $22/7$ use karo, warna 3.14.

3. Area vs Perimeter

EXAM * Exam me CONSTANTLY aata hai

- Area = andar ki jagah (unit: cm^2).
- Perimeter / circumference = boundary ki length (unit: cm).

4. Circle ke 2 Formula

EXAM * Exam me CONSTANTLY aata hai

- Circumference = $2\pi r$.
- Area = πr^2 .

5. Angle (degree)

EXAM * Exam me CONSTANTLY aata hai

- Pura circle = 360 degree.
- Sector ek hissa hai jiska apna angle theta hota hai.

6. Fraction of Circle

EXAM * Exam me CONSTANTLY aata hai

- Sector = circle ka $\theta/360$ hissa.
- Isi se sector ka area aur arc length nikalte hain.

7. Square / Units

- $r^2 = r \times r$.

- Area ka unit hamesha square hota hai (cm^2).

PART B - Chapter 11 ka Main Content

8. Circumference aur Area of Circle

EXAM * Exam me CONSTANTLY aata hai

Circumference = $2\pi r$

Area = πr^2

$\pi = 22/7$ (ya 3.14).

- Example: $r=7 \rightarrow$ Circumference = $2 \times 22/7 \times 7 = 44 \text{ cm}$.

9. Length of an Arc

EXAM * Exam me CONSTANTLY aata hai

Arc length = $(\theta / 360) \times 2\pi r$

θ = sector ka angle (degree me).

- Yani poori circumference ka $\theta/360$ hissa.

10. Area of a Sector

EXAM * Exam me CONSTANTLY aata hai

Area of sector = $(\theta / 360) \times \pi r^2$

= poore circle ke area ka $\theta/360$ hissa.

- Minor sector = chhota hissa; Major sector = bada hissa.

11. Area of a Segment

EXAM * Exam me CONSTANTLY aata hai

Area of segment = Area of sector - Area of triangle

(triangle = do radii aur chord se banta hai).

- Minor segment chord ke ek taraf, major segment doosri taraf hota hai.

12. Combination / Shaded Area

- Shaded area = badi shape ka area - chhoti shape ka area.
- Jaise square me circle, ya do circles ka antar (ring).
- Ring (annulus) area = $\pi(R^2 - r^2)$.

Quick Revision

Circumference = $2 \pi r$; Area = πr^2 .

Arc length = $(\theta/360) \times 2 \pi r$.

Sector area = $(\theta/360) \times \pi r^2$.

Segment = sector - triangle.

$\pi = 22/7$ (r me 7 ho) warna 3.14.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: $r = 7$ cm. Area of circle? ($\pi = 22/7$)

- Step: Area = $\pi r^2 = (22/7) \times 7 \times 7$.
- Step: = 22×7 .

Answer: Area = 154 cm^2

Example 2: $r = 7$ cm. Circumference? ($\pi = 22/7$)

- Step: Circumference = $2 \pi r = 2 \times (22/7) \times 7$.
- Step: = 2×22 .

Answer: 44 cm

Example 3: $r = 14$ cm. Area? ($\pi = 22/7$)

- Step: Area = $(22/7) \times 14 \times 14 = 22 \times 2 \times 14$.
- Step: = 22×28 .

Answer: 616 cm^2

Example 4: Sector $\theta=90$, $r=14$. Area? ($\pi=22/7$)

- Step: Sector area = $(90/360) \times \pi r^2 = (1/4) \times (22/7) \times 196$.
- Step: = $(1/4) \times 616$.

Answer: 154 cm^2

Example 5: Arc length, $\theta=60$, $r=21$. ($\pi=22/7$)

- Step: Arc = $(60/360) \times 2 \pi r = (1/6) \times 2 \times (22/7) \times 21$.
- Step: = $(1/6) \times 132$.

Answer: 22 cm

Example 6: Diameter = 28 cm. Radius aur area? ($\pi=22/7$)

- Step: $r = 28/2 = 14$. Area = $(22/7) \times 14 \times 14$.
- Step: = 616.

Answer: $r=14$, Area = 616 cm^2

Example 7: Sector $\theta=180$ (half), $r=7$. Area? ($\pi=22/7$)

- Step: $(180/360) \times \pi r^2 = (1/2) \times (22/7) \times 49$.
- Step: = $(1/2) \times 154$.

Answer: 77 cm^2

Example 8: Ring: $R=7$, $r=\text{conf...}$ $r = \text{inner}$ 0? Outer 7 area.

- Step: Area of circle = $\pi R^2 = (22/7) \times 49$.
- Step: = 154.

Answer: 154 cm^2

Example 9: $\theta=90$, $r=10$. Arc length? ($\pi=3.14$)

- Step: Arc = $(90/360) \times 2 \times 3.14 \times 10 = (1/4) \times 62.8$.

- Step: = 15.7.

Answer:15.7 cm

Example 10: Quarter circle area, $r=4$. ($\pi=3.14$)

- Step: Quarter = $(1/4) \pi r^2 = (1/4) \times 3.14 \times 16$.
- Step: = $(1/4) \times 50.24$.

Answer:12.56 cm^2

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1.** Circle ke area ka formula kya hai?
- Q2.** Circle ke circumference ka formula kya hai?
- Q3.** π ka value kya hota hai (2 forms)?
- Q4.** Pura circle kitne degree ka hota hai?
- Q5.** Sector ka area formula kya hai?
- Q6.** Arc length ka formula kya hai?
- Q7.** Segment ka area kaise nikalte hain?
- Q8.** $r=7$, $\pi=22/7$ -> area?
- Q9.** $r=7$, $\pi=22/7$ -> circumference?
- Q10.** Diameter=14 ho to radius?
- Q11.** Sector $\theta=90$, $r=14$ -> area? ($\pi=22/7$)
- Q12.** Area ka unit kya hota hai?
- Q13.** Perimeter/circumference ka unit kya hota hai?
- Q14.** $\theta=180$ ka sector circle ka kitna hissa hai?
- Q15.** Ring (annulus) ka area formula?
- Q16.** $r=14$, $\pi=22/7$ -> area?
- Q17.** Half circle ($r=7$) ka area? ($\pi=22/7$)
- Q18.** Sector circle ka kitna fraction hota hai?
- Q19.** Quarter circle ka angle kitna hota hai?
- Q20.** Shaded area kaise nikalte hain (general rule)?

Step-by-Step Solutions

Q1. Circle area formula?

- Step: Standard formula.

Answer: πr^2

Q2. Circumference formula?

- Step: Standard formula.

Answer: $2 \pi r$

Q3. π ka value?

- Step: Do common forms.

Answer:22/7 ya 3.14

Q4. Pura circle kitne degree?

- Step: Full angle.

Answer:360 degree

Q5. Sector area formula?

- Step: $\theta/360$ hissa of πr^2 .

Answer: $(\theta/360) \times \pi r^2$

Q6. Arc length formula?

- Step: $\theta/360$ hissa of $2 \pi r$.

Answer: $(\theta/360) \times 2 \pi r$

Q7. Segment area?

- Step: Sector me se triangle ghatao.

Answer:Sector area - triangle area

Q8. $r=7$, $\pi=22/7$ -> area?

- Step: $(22/7) \times 49 = 22 \times 7$.

Answer:154 cm^2

Q9. $r=7$ -> circumference?

- Step: $2 \times (22/7) \times 7 = 44$.

Answer:44 cm

Q10. Diameter=14 -> radius?

- Step: $r = d/2$.

Answer:7

Q11. Sector $\theta=90$, $r=14$ -> area?

- Step: $(1/4) \times (22/7) \times 196 = (1/4) \times 616$.

Answer:154 cm^2

Q12. Area ka unit?

- Step: 2-D jagah.

Answer: cm^2 (square)

Q13. Circumference ka unit?

- Step: Length.

Answer:cm

Q14. $\theta=180$ sector hissa?

- Step: $180/360 = 1/2$.

Answer:Aadha (half)

Q15. Ring area formula?

- Step: Bahar - andar.

Answer: $\pi(R^2 - r^2)$

Q16. $r=14$, $\pi=22/7$ -> area?

- Step: $(22/7) \times 196 = 22 \times 28$.

Answer:616 cm^2

Q17. Half circle $r=7$ area?

- Step: $(1/2) \times (22/7) \times 49 = (1/2) \times 154$.

Answer:77 cm^2

Q18. Sector fraction?

- Step: Angle ratio.

Answer: $\theta/360$

Q19. Quarter circle angle?

- Step: $360/4$.

Answer: 90 degree

Q20. Shaded area rule?

- Step: Bade me se chhota ghatao.

Answer: Badi shape - chhoti shape